
OmniVoice: Towards Omnilingual Zero-Shot Text-to-Speech with Diffusion Language Models

Han Zhu, Lingxuan Ye, Wei Kang, Zengwei Yao, Liyong Guo, Fangjun Kuang
Zhifeng Han, Weiji Zhuang, Long Lin, Daniel Povey
Xiaomi Corp., China
{zhuhan3, dpovey}@xiaomi.com

Abstract

We present OmniVoice, a massive multilingual zero-shot text-to-speech (TTS) model that scales to over 600 languages. At its core is a novel diffusion language model-style discrete non-autoregressive (NAR) architecture. Unlike conventional discrete NAR models that suffer from performance bottlenecks in complex two-stage (text-to-semantic-to-acoustic) pipelines, OmniVoice directly maps text to multi-codebook acoustic tokens. This simplified approach is facilitated by two key technical innovations: (1) a full-codebook random masking strategy for efficient training, and (2) initialization from a pre-trained LLM to ensure superior intelligibility. By leveraging a 581k-hour multilingual dataset curated entirely from open-source data, OmniVoice achieves the broadest language coverage to date and delivers state-of-the-art performance across Chinese, English, and diverse multilingual benchmarks. Our code and pre-trained models are publicly available¹.

1 Introduction

Zero-shot text-to-speech (TTS) models trained on large-scale datasets have demonstrated a remarkable ability to generate high-quality speech conditioned on only a few seconds of reference audio [1, 2]. Despite these advances, most existing models support only a limited set of languages, often leaving hundreds of low-resource languages behind. Expanding language coverage is not merely a technical challenge but also a crucial step toward extending speech technologies to languages across the globe. To address this gap, we aim to develop a massive multilingual zero-shot TTS model supporting hundreds of languages. Realizing such extensive scaling requires a TTS architecture with exceptional capacity and robustness to handle diverse linguistic patterns.

In the pursuit of such scalable and high-quality TTS, current research primarily adheres to two paradigms: autoregressive (AR) [3, 4, 5, 6, 7, 8, 9, 10, 11] and non-autoregressive (NAR) models [12, 13, 14]. NAR models offer advantages in both inference speed (via parallel decoding) and robustness (owing to bidirectional context) [12, 14, 15]. Within NAR frameworks, models can be broadly categorized into continuous-latent-based [14, 16, 17, 18] and discrete-token-based [19]. The latter was shown to yield superior prosodic diversity [20].

However, state-of-the-art (SOTA) discrete-token NAR systems [19] typically rely on complex two-stage cascaded pipelines (text-to-semantic followed by semantic-to-acoustic). While such decoupling simplifies the training of individual modules, it also introduces significant drawbacks: (1) error propagation, where inaccuracies in semantic prediction degrade the final audio quality; and (2) information bottlenecks, where low-bitrate semantic representations sacrifice fine-grained acoustic details. While single-stage alternatives [21] attempt to bypass these issues, they have historically lagged behind two-stage systems in terms of speech intelligibility.

¹<https://github.com/k2-fsa/OmniVoice>

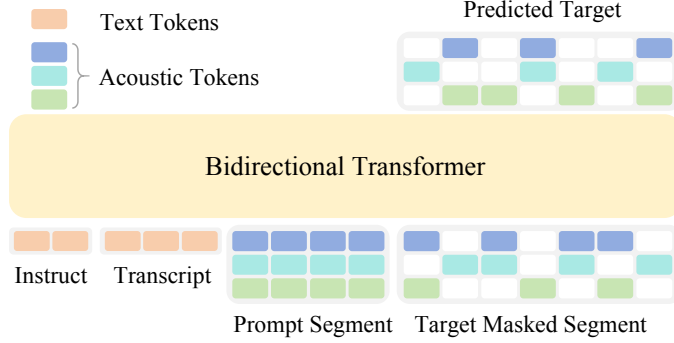


Figure 1: Illustration of OmniVoice architecture.

To bridge this gap, we introduce OmniVoice, an architecturally streamlined yet highly effective discrete NAR TTS framework. OmniVoice employs a discrete masked diffusion objective [22] with a bidirectional Transformer [23] to directly map text to multi-codebook acoustic tokens, thereby bypassing the complexity and limitations of cascaded pipelines. Its core modeling philosophy extends the success of diffusion language models [24, 25] to the speech domain. We demonstrate that the potential of this minimalist architecture can be fully unleashed through two technical innovations:

(1) Full-Codebook Random Masking: Conventional multi-codebook acoustic prediction methods adopt "per-layer" masking schedules [26, 19], suffering from inefficient convergence. We propose a fully stochastic masking strategy across all codebooks that significantly enhances training efficiency and generative quality.

(2) LLM Initialization: To resolve the intelligibility issues in single-stage discrete NAR models, we initialize our backbone with pre-trained AR LLM weights to inherit linguistic knowledge, making OmniVoice the first NAR TTS model to successfully benefit from LLM initialization [3].

The architectural advantage of OmniVoice makes it uniquely suited for scaling across diverse linguistic contexts. We curated a 581k-hour multilingual dataset encompassing more than 600 languages, derived exclusively from open-source resources. Training on this dataset allows OmniVoice to achieve the most extensive language coverage reported to date, bridging the gap for hundreds of previously under-served low-resource languages.

Beyond its extensive language coverage, OmniVoice supports multi-dimensional controllability, including prompt denoising [27, 28], speaker attribute-based voice design [29], and fine-grained paralinguistic and phonetic control [30, 31]. These features significantly enhance its versatility, making OmniVoice well-suited for a wide range of real-world applications.

Comprehensive experimental evaluations across Chinese, English, and massive multilingual benchmarks (covering up to 102 languages) validate that OmniVoice delivers SOTA performance in intelligibility, speaker similarity and naturalness, setting a new frontier for high-quality, high-coverage multilingual TTS.

2 Proposed Method

This section elaborates the design of OmniVoice. We first introduce the streamlined single-stage architecture that serves as the high-performance backbone for massive multilingual modeling, followed by two core technical optimizations to boost model training efficiency and speech intelligibility. On this basis, we further present the large-scale multilingual data construction and balancing strategies to realize extensive language coverage, and finally equip the model with multi-dimensional controllability for practical deployment.

2.1 Architecture

OmniVoice is a single-stage NAR TTS model with a diffusion language model style architecture [24, 25]. Specifically, the model is trained with discrete diffusion objection and adopts a bidirectional Transformer backbone. OmniVoice directly maps text to multi-codebook acoustic tokens,

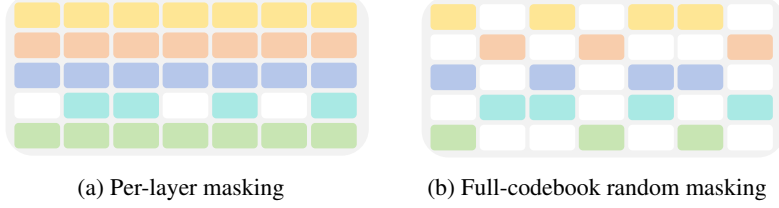


Figure 2: Comparison of per-layer masking and full-codebook random masking. The x-axis denotes the time dimension, the y-axis the codebook dimension, and white blocks indicate masked tokens.

eliminating the error propagation and information bottlenecks issues in conventional two-stage cascaded pipelines. This end-to-end streamlined architecture lays the foundation for large-scale multilingual modeling, with targeted optimizations and scaling strategies detailed in the subsequent sections.

The architecture of OmniVoice is illustrated in Fig. 1. The input of OmniVoice comprises two components:

- Text token sequence (Y): A concatenated sequence of instruct and transcript tokens, providing the linguistic and task-oriented guidance.
- Acoustic token matrix (X): A multi-codebook matrix $X \in \mathbb{R}^{T \times C}$, where T is the number of time steps and C is the number of codebooks.

The acoustic matrix X is partitioned along the temporal dimension into two segments: the prompt segment X_{prompt} , which contains the prefix acoustic context, and the target masked segment X_{prompt} , where tokens are randomly replaced with a special mask token $[M]$. The model is designed to leverage the text conditions Y , the prompt X_{prompt} and unmasked tokens in X_{prompt} to recover the original tokens in the masked position in X_{prompt} .

The text tokens are embedded via a text embedding layer, and the acoustic tokens are embedded via codebook-specific embedding layers. The embeddings of all C codebooks at the same temporal position are summed to form a unified embedding, which is then fed into a bidirectional Transformer. On the output side, to reconstruct the multi-codebook tokens, OmniVoice employs C independent, codebook-specific prediction heads. Each head projects the final hidden states to output a probability distribution over the vocabulary of its corresponding codebook.

The training loss is computed on the masked acoustic token positions, aiming to optimize the model for accurate token recovery. Let $\mathcal{M}$ denote the set of indices (t, c) corresponding to masked positions within the target segment, where $t \in \{T_p + 1, \dots, T\}$ and $c \in \{1, \dots, C\}$. The training loss $\mathcal{L}$ is formulated as:

$$\mathcal{L} = - \sum_{(t,c) \in \mathcal{M}} \log P(x_{t,c} | X, Y; \theta) \quad (1)$$

where $x_{t,c}$ is the ground-truth acoustic token at time step t and codebook index c , and $P(x_{t,c} | \dots; \theta)$ is the probability distribution predicted by the model parameterized by θ .

2.1.1 Full-Codebook Random Masking for Training Efficiency

Masking design is critical for OmniVoice, as it directly determines the training pattern and loss function. Previous methods for multi-codebook acoustic token prediction typically employ a "per-layer" masking schedule [26, 19], which randomly masks within a single codebook layer c per sample and computes the loss exclusively for that layer. Such masking strategy is designed to align with the layer-wise inference. However, it optimizes only a sparse subset of the token matrix at each iteration, leading to suboptimal training efficiency.

To circumvent this limitation, OmniVoice adopts a fully stochastic masking strategy across all codebook layers. Specifically, we independently sample a binary mask $m_{i,j} \sim \text{Bernoulli}(p_t)$ for every entry in the $T \times C$ token matrix, where the masking ratio p_t is drawn from a uniform distribution $p_t \sim \mathcal{U}(0, 1)$ for each training instance. Consequently, on average, 50% of the tokens are used for

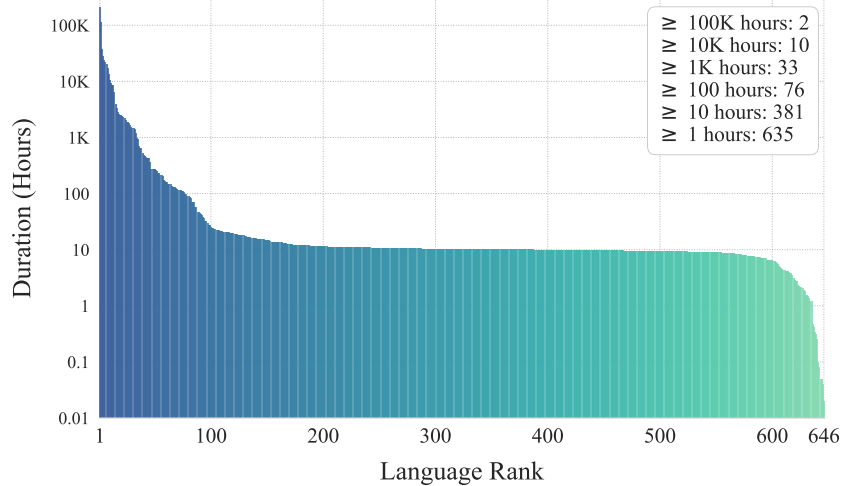


Figure 3: Statistics of the multilingual training dataset.

loss computation, C times more than per-layer masking strategy, significantly accelerating convergence and boosting generative quality.

2.1.2 LLM Initialization for Intelligibility

Existing discrete NAR TTS models [19, 21], particularly single-stage architectures [21], often exhibit suboptimal intelligibility compared to their counterparts. To address this, we propose a simple yet effective strategy: initializing the model backbone with pre-trained AR large language models (LLMs) [32]. While LLM-based initialization has been successfully integrated into recent AR-TTS frameworks [3, 33], OmniVoice is the first NAR TTS model that successfully leverages LLM initialization to achieve notable intelligibility gains. Our backbone is structurally identical to standard AR LLMs, facilitating direct weight transfer. While these LLMs are originally trained with a causal mask, we empirically find that their pre-trained knowledge translates well to our bidirectional architecture. This initialization allows OmniVoice to repurpose LLM’s linguistic capacity as a strong prior for text-to-speech mapping, significantly enhancing the intelligibility of the generated speech.

2.2 Multilingual Scaling

A core objective of this work is to extend language support to over 600 languages, including hundreds of low-resource languages that are rarely supported in mainstream TTS systems. OmniVoice’s streamlined end-to-end architecture is inherently well-suited for such large-scale multilingual scaling.

Expanding language coverage of TTS is a long-standing challenge [34, 35, 36, 37, 38]. While MMS [39] scales to more than 1,000 languages, yet lacks zero-shot voice cloning capabilities and relies on language-specific modeling. Conversely, current multilingual zero-shot TTS models remain limited to narrow linguistic scopes, covering only a few dozen languages [40, 41, 36, 42, 33, 43, 38]. To bridge this gap, we introduce the first massively multilingual zero-shot TTS model capable of generalizing across hundreds of diverse languages within a single, unified model.

Data acquisition remains one of the primary bottlenecks for large-scale multilingual expansion. To address this, we aggregated 50 datasets by leveraging open-source community efforts (see Appendix A for the complete list). Given the heterogeneous quality of these sources (many of which were not originally designed for TTS and contain noisy audio or transcriptions), we employed a speech restoration model [44] to enhance degraded speech and applied rule-based filtering [17] to exclude invalid transcriptions. The resulting corpus comprises 581k hours of audio spanning more than 600 languages (detailed statistics in Fig. 3). Training on this unprecedented dataset empowers OmniVoice to achieve the broadest language coverage reported to date, significantly advancing the SOTA in multilingual TTS.

To mitigate the severe data imbalance inherent in massively multilingual datasets, where high-resource languages vastly outnumber low-resource ones, we apply a language-level data resampling strategy. Specifically, we upsample the training data of low-resource languages by assigning a repetition factor, r_i , to each language i . Let D_i denote the total audio duration for language i , and $D_{\max}$ represent the maximum duration across all languages. The repetition factor is formulated as:

$$r_i = \max \left(1, \text{round} \left(\left(\frac{D_{\max}}{D_i} \right)^{1-\beta} \right) \right) \quad (2)$$

where $\beta \in [0, 1]$ is a hyperparameter controlling the degree of smoothing. Setting $\beta = 1.0$ retains the natural long-tail distribution of the dataset, whereas $\beta = 0.0$ yields a uniform distribution across all languages. In our implementation, we empirically set $\beta = 0.8$ to keep the performance of high-resource languages while ensuring adequate model exposure to low-resource languages.

For multilingual text processing, we employ the subword tokenizer of pre-trained LLMs, eliminating cumbersome grapheme-to-phoneme conversion and language-specific text normalization. This strategy also allows the LLM backbone to inherit and leverage its pre-trained knowledge efficiently.

2.3 Multi-Dimensional Controllability

Beyond its extensive multilingual coverage, we further equip OmniVoice with multi-dimensional controllability across acoustic, identity, and linguistic aspects. This comprehensive control mechanism significantly improves the model’s practicality for complex, real-world applications.

2.3.1 Acoustic Control: Prompt Denoising

In practical scenarios, audio prompts are often recorded in non-ideal conditions, potentially compromised by environmental noise or reverberation. To prevent the model from replicating these unwanted artifacts, we implement a prompt denoising task [27, 28]. During training, we augment a subset of the data by injecting synthetic noise and reverberation into the prompt segments. These samples are paired with a specific instruction token, `<|denoise|>`. This mechanism compels the model to disentangle the speaker’s intrinsic voice identity from the acoustic environment, enabling the synthesis of clean, high-fidelity speech even when conditioned on degraded prompts.

2.3.2 Identity Control: Speaker-Attribute-Based Voice Design

To enable flexible TTS in the absence of an audio prompt, OmniVoice supports speaker-attribute-based voice design [29, 42]. By incorporating specific speaker attributes (e.g., gender, age, pitch, and accent/dialect) into the training instruction sequence, the model can synthesize highly customized voices on demand.

2.3.3 Linguistic Control: Paralinguistics and Phonetics

To bridge the gap between basic intelligibility and human-like expressiveness, we incorporate paralinguistic control (e.g., laughter) by training on datasets enriched with affective cues [30, 45].

Furthermore, recognizing that models may occasionally encounter challenges with the pronunciation of linguistic corner cases, such as Chinese polyphonic characters or specialized English terminology, we adopt a hybrid text input format [31] to provide users with explicit phonetic override capabilities. During training, we stochastically replace characters or words with their corresponding phonetic transcriptions, specifically Pinyin for Chinese and phonemes from the CMU pronunciation dictionary for English. This mechanism allows for deterministic control over pronunciation during inference, enabling the model to handle complex linguistic scenarios with high precision.

3 Experimental Setup

This section details the training datasets, model configurations, training and inference protocols, evaluation benchmarks, and evaluation metrics of our experiments.

3.1 Training Datasets

We train OmniVoice under two distinct data configurations. First, a bilingual variant is trained on the Chinese and English subsets of the Emilia dataset [46], enabling fair comparison with existing SOTA zero-shot TTS models trained on the same data. Prompt denoising is omitted in this variant to isolate its impact and highlight the inherent advantages of our architecture. Second, a multilingual variant is trained on a self-built 581k-hour multilingual dataset spanning 600+ languages.

3.2 Model Details

OmniVoice is built on a bidirectional Transformer backbone, which is initialized with the pre-trained LLM weights of Qwen3-0.6B [32]. The Higgs-audio tokenizer [47] is adopted to extract 8-codebook acoustic tokens and reconstruct audio from these tokens.

3.3 Training

The AdamW optimizer [48] is used, with a peak learning rate of $1e-4$ and a cosine learning rate schedule that includes 3% of total training steps as warmup. Mixed precision (BF16) and sequence packing (8192 tokens per GPU) are employed during training to improve efficiency. Using 8 H800 GPUs, the multilingual variant (2M training updates) and the bilingual Emilia variant (300k training updates) were trained in 9.66 days and 1.33 days, respectively.

3.4 Inference

During inference, we perform a 32-steps iterative unmasking process. The cumulative proportion of unmasked tokens at step n , denoted as r_n , follows a time-shifted schedule:

$$r_n = \frac{\tau \cdot (n/N)}{1 + (\tau - 1) \cdot (n/N)} \quad (3)$$

where $N = 32$ is the total number of steps, and $\tau = 0.1$ is the shift parameter. The proportion of tokens to be newly unmasked at each step n is given by $k_n = r_n - r_{n-1}$, where $r_0 = 0$.

The selection of unmasking positions and token identities is handled as follows:

- **Position Selection:** At each step, we identify k_n positions to be unmasked by sampling from the confidence scores in the log-softmax space. To introduce beneficial stochasticity and avoid local optima, we apply a temperature $T = 5$ to the confidence scores before sampling the indices.
- **Token Assignment:** Once the positions are selected, the specific token class for each position is determined deterministically by taking the *argmax* of the predicted probability distribution.

We also apply a layer penalty on the confidence scores to encourage unmasking lower-layer tokens first. Furthermore, classifier-free guidance [49] is utilized in the log-softmax space with a guidance scale of 2. All aforementioned strategies are verified to effectively improve model performance and generation stability.

3.4.1 Evaluation Benchmarks

We evaluate OmniVoice on four benchmarks covering standard Chinese/English settings and massive multilingual scenarios:

- **LibriSpeech-PC [50, 14]:** A standard English zero-shot TTS benchmark.
- **Seed-TTS [2]:** A bilingual (Chinese/English) zero-shot benchmark.
- **MiniMax-Multilingual-24:** A multilingual benchmark covering 24 languages, introduced in [51].
- **FLEURS-Multilingual-102:** A 102-language multilingual benchmark constructed from the dev/test splits of the FLEURS dataset [52] to further evaluate OmniVoice’s multilingual capability. It is the zero-shot TTS evaluation benchmark with the widest language coverage to date.

3.5 Evaluation Metrics

Our evaluation combines several objective and subjective metrics. For speaker similarity evaluation, we use SIM-o [12] with a WavLM-based [53] ECAPA-TDNN model [54]. Intelligibility is measured using word error rate (WER) or character error rate (CER) across different languages. Specifically, we use the Hubert-based ASR model [55] for LibriSpeech-PC test-clean, Paraformer-zh [56] for Chinese, the Omnilingual ASR model [57] for the FLEURS benchmark, and Whisper-large-v3 [58] for the remaining datasets. We also adopt UTMOS [59] to assess objective speech naturalness.

These objective metrics are supplemented with subjective evaluations, including comparative mean opinion score (CMOS, $[-3, 3]$) and similarity mean opinion score (SMOS, $[0, 5]$), which measure human opinions on relative speech quality and absolute speaker similarity to the prompt audio.

4 Experimental Results

4.1 Evaluation on English and Chinese

Table 1: Objective evaluation results on Chinese and English test sets. All baseline results are obtained using official checkpoints. Params. (Parameters) denotes the total parameter size of voice cloning TTS systems (including audio tokenizer, vocoder, and other related components). Best results are highlighted in **bold**. Top: Results on LibriSpeech-PC. Bottom: Results on Seed-TTS test sets.

Model	Params.	Training Data (hours)	LibriSpeech-PC test-clean		
			SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
Ground-truth	-	-	0.690	1.87	4.10
AR Models					
IndexTTS2	1.7B	55k Multi.	0.700	2.35	4.06
CosyVoice3	1.1B	1000k Multi.	0.694	1.59	4.28
VoxCPM	0.7B	1800k Multi.	0.717	1.74	4.18
Qwen3-TTS	1.1B	5000k Multi.	0.704	1.60	4.41
NAR Models					
F5-TTS	0.4B	100K Emilia	0.655	1.89	3.89
ZipVoice	0.1B	100k Emilia	0.668	1.64	3.98
MaskGCT	2.2B	100K Emilia	0.691	2.26	3.91
OmniVoice-Emilia	0.8B	100k Emilia	0.697	1.57	4.23
OmniVoice	0.8B	581k Multi.	0.729	1.30	4.28

Model	Seed-TTS test-en			Seed-TTS test-zh		
	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
Ground-truth	0.734	2.14	3.52	0.755	1.25	2.78
AR Models						
IndexTTS2	0.706	2.33	3.65	0.764	1.05	3.00
CosyVoice3	0.696	2.17	3.96	0.778	1.14	3.32
VoxCPM	0.731	1.92	3.77	0.772	0.99	2.94
Qwen3-TTS	0.708	1.54	4.16	0.766	1.15	3.46
NAR Models						
F5-TTS	0.664	1.85	3.72	0.750	1.53	2.93
ZipVoice	0.697	1.70	3.82	0.751	1.40	3.15
MaskGCT	0.713	2.88	3.55	0.773	2.40	2.63
OmniVoice-Emilia	0.717	1.72	3.88	0.765	0.89	3.05
OmniVoice	0.741	1.60	3.91	0.777	0.84	3.11

Table 2: Subjective evaluation results on Chinese and English test sets. CMOS and SMOS are used for evaluation. Best results are highlighted in **bold**.

Model	CMOS $\uparrow$	SMOS $\uparrow$
Ground-truth	0.00	3.02 $\pm$ 0.20
Qwen3-TTS	0.40 $\pm$ 0.16	3.65 $\pm$ 0.18
ZipVoice	-0.30 $\pm$ 0.16	3.35 $\pm$ 0.19
MaskGCT	-0.38 $\pm$ 0.17	3.20 $\pm$ 0.18
OmniVoice-Emilia	0.42 $\pm$ 0.15	3.58 $\pm$ 0.18
OmniVoice	0.44 $\pm$ 0.16	3.80 $\pm$ 0.17

Table 1 and Table 2 summarize the Chinese and English performance of OmniVoice in comparison with SOTA AR/NAR TTS models.

OmniVoice-Emilia surpasses all NAR baselines (F5-TTS [14], ZipVoice [16], MaskGCT [19]) trained on the same Emilia corpus, verifying the effectiveness of our proposed architecture.

The final multilingual version OmniVoice model yields competitive overall performance across all benchmarks against baselines trained on unconstrained datasets (IndexTTS2 [5], CosyVoice3 [33], VoxCPM [10], Qwen3-TTS [42]), with particular advantages in speaker similarity and intelligibility. This demonstrates OmniVoice’s strong capability on the two most high-resource languages.

4.2 Evaluation on Multilingual Benchmarks

We validate OmniVoice’s multilingual capability on the 24-language MiniMax-Multilingual-24 benchmark and the 102-language FLEURS-Multilingual-102 benchmark.

Table 3: Evaluation on the MiniMax-Multilingual-24 test set.

Language	WER $\downarrow$			SIM-o $\uparrow$		
	OmniVoice	MiniMax	ElevenLabs	OmniVoice	MiniMax	ElevenLabs
Arabic	1.392	1.665	1.666	0.776	0.736	0.706
Cantonese	17.709*	34.111	51.513	0.838	0.778	0.670
Chinese	1.008	2.252	16.026	0.821	0.780	0.677
Czech	2.856	3.875	2.108	0.837	0.796	0.685
Dutch	1.358	1.143	0.803	0.813	0.738	0.680
English	1.560	2.164	2.339	0.884	0.756	0.613
Finnish	3.750	4.666	2.964	0.864	0.835	0.759
French	3.347	4.099	5.216	0.801	0.628	0.535
German	0.964	1.906	0.572	0.812	0.733	0.614
Greek	1.057	2.016	0.991	0.867	0.826	0.733
Hindi	4.330	6.962	5.827	0.863	0.818	0.730
Indonesian	1.973	1.237	1.059	0.805	0.729	0.660
Italian	2.070	1.543	1.743	0.812	0.699	0.579
Japanese	4.027	3.519	10.646	0.828	0.776	0.738
Korean	2.651	1.747	1.865	0.828	0.776	0.700
Polish	0.874	1.415	0.766	0.877	0.802	0.729
Portuguese	2.511	1.877	1.331	0.859	0.805	0.711
Romanian	2.424	2.878	1.347	0.836	0.809	0.699
Russian	2.233	4.281	3.878	0.783	0.761	0.676
Spanish	1.026	1.029	1.084	0.804	0.762	0.615
Thai	3.978	2.701	73.936	0.841	0.800	0.588
Turkish	2.166	1.520	0.699	0.851	0.779	0.596
Ukrainian	1.774	1.082	0.997	0.810	0.730	0.647
Vietnamese	1.373	0.880	73.415	0.804	0.743	0.369
Average	2.850	3.774	10.950	0.830	0.766	0.655

* Cantonese WER is 2.273% when evaluated with the SenseVoice-Small ASR model.

As shown in Table 3, despite being trained exclusively on open-source datasets, OmniVoice clearly outperforms leading commercial systems (ElevenLabs v2 and MiniMax-Speech) in both average SIM-o and WER, demonstrating that OmniVoice achieves commercial-grade multilingual TTS performance. While Cantonese shows a higher WER in the table, our analysis reveals that this is due to limitations in the Whisper ASR model rather than the quality of the generated speech. When evaluated with the SenseVoice-Small ASR model [60], the WER for Cantonese decreases to 2.273%. However, to maintain consistency with the benchmarks in [51], we retain the Whisper-based results in our report.

Table 4: Evaluation on the FLEURS-Multilingual-102 test set.

Model	FLEURS-Multilingual-102			
	Avg SIM-o $\uparrow$	Avg CER $\downarrow$	Languages with CER $\leq$	
			5%	10%
Ground-truth	-	5.11	75	92
OmniVoice	0.788	4.00	82	95

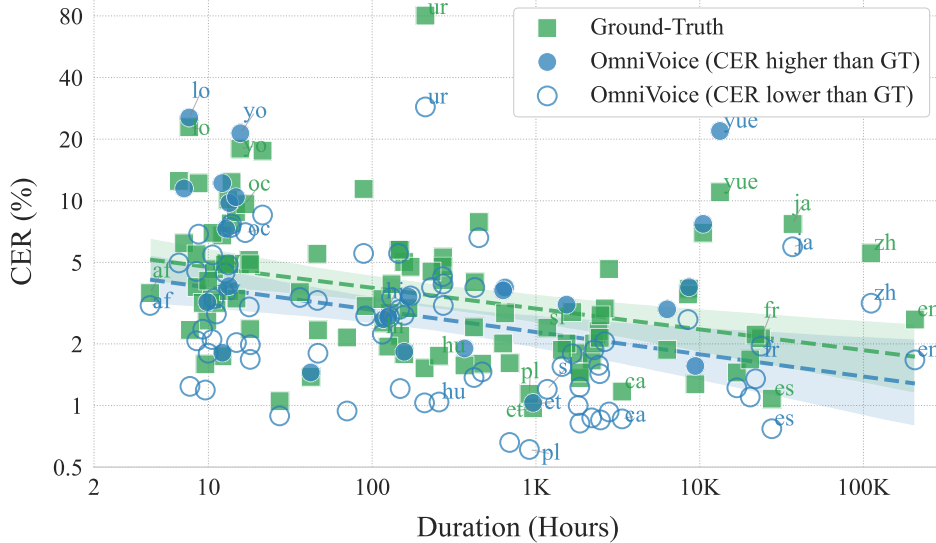


Figure 4: CERs of OmniVoice vs. ground truth across languages with varying training data durations on FLEURS-Multilingual-102. Hollow circles indicate languages where OmniVoice achieves lower CER than ground-truth, Filled circles indicate higher CER.

Next, we evaluate OmniVoice on our self-built FLEURS-Multilingual-102 benchmark. We report the average SIM-o, average CER, and the number of languages below specific CER thresholds in Table 4. We observe that OmniVoice achieves an average CER of 4.00%, which is comparable to that of the ground truth. Per-language CER results are provided in Appendix C.

To better understand OmniVoice’s performance, we plot a figure illustrating the relationship between per-language CER and training data duration in Fig. 4. It can be seen that OmniVoice maintains high intelligibility (CER < 5%) even for many languages with less than 10 hours of training data, demonstrating strong generalization capability on low-resource languages. Note that we do not claim OmniVoice can generate speech of better quality than the ground truth for all languages. However, OmniVoice’s performance has exceeded the measurement capability of existing ASR models.

4.3 Effectiveness of Key Designs

To validate the effectiveness of our core designs, we conduct ablation experiments on the Emilia dataset. All models are trained for 300k updates with the same hyperparameters, with prompt denoising disabled unless otherwise specified.

Table 5: Impact of different masking strategies.

Masking Strategy	Librispeech-PC test-clean		
	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
SoundStorm-style mask	0.661	3.00	4.12
MaskGCT-style mask	0.660	2.04	4.17
Full-codebook random mask	0.697	1.57	4.23
- compute loss on a single codebook	0.648	2.85	4.22

Firstly, we compare different masking strategies to validate the effectiveness of our proposed full-codebook random masking strategy. SoundStorm-style masking is a variant of the per-layer masking strategy, which samples layers for loss computation from a uniform distribution and selects the masking ratio according to a cosine function. MaskGCT-style masking further samples layers based on a linear distribution, where lower layers are selected with higher probability.

As shown in Table 5, full-codebook random masking consistently outperforms the single-codebook masking strategies adopted by MaskGCT and SoundStorm. Furthermore, we conduct an experiment where the loss is calculated only on a single codebook, which also results in significant performance degradation, confirming the advantages of the proposed dense loss computation.

Table 6: Impact of LLM initialization on WER across datasets. Abbreviations: Libri = Librispeech-PC test-clean, Seed-en = Seed-TTS test-en, Seed-zh = Seed-TTS test-zh.

Model	Learning Rate	WER $\downarrow$		
		Libri	Seed-en	Seed-zh
LLM initialization	1e-4	1.57	1.72	0.89
Random initialization	1e-4	2.79	2.34	1.11
	2e-4	2.52	2.43	1.01
	5e-4	2.56	2.07	1.01
	1e-3	2.72	2.29	1.02

We illustrate the importance of LLM initialization for high intelligibility (i.e., low WER) in Table 6. It can be observed that even after extensive learning rate tuning, the WERs of models without LLM initialization are still consistently higher than those of the model with LLM initialization, highlighting the importance of inheriting linguistic knowledge from pre-trained LLMs.

Table 7: Impact of prompt denoising task.

Model	Librispeech-PC test-clean		
	SIM-o $\uparrow$	WER $\downarrow$	UTMOS $\uparrow$
w/o prompt denoise	0.697	1.57	4.23
w/ prompt denoise	0.668	1.56	4.32

We illustrate the impact of prompt denoising in Table 7. When prompt denoising is enabled, UTMOS improves from 4.23 to 4.32 because the model generates cleaner speech, while SIM-o decreases slightly ($0.697 \rightarrow 0.668$) because the generated speech is more standardized than the noisy prompt, which aligns with our design objective of generating high-quality speech from degraded prompts.

4.3.1 Inference Speed

We evaluate the inference speed of OmniVoice on an H20 GPU using PyTorch as the inference framework. Following the identical experimental setup in [16, 14], we use a 3-second audio prompt to generate 10-second audio clips, and report the real-time factor (RTF) across different batch sizes and inference steps. As shown in Table 8, OmniVoice achieves an RTF of 0.0319 with 16 inference

steps and a batch size of 1, outperforming the corresponding results of ZipVoice (0.0557) with the same inference step configuration. Notably, OmniVoice also yields better generation quality in the same 16-step setting (shown in Appendix B). When batch inference is employed, OmniVoice attains an even lower RTF of 0.022. In addition, its inference efficiency can be further improved with acceleration techniques such as TensorRT. These results demonstrate the high inference efficiency of OmniVoice.

Table 8: RTF with different inference steps and batch sizes.

Inference Steps	Batch Size			
	1	2	4	8
16	0.0319	0.0263	0.0235	0.0224
32	0.0598	0.0486	0.0436	0.0414

5 Conclusions

In this work, we introduce OmniVoice, a massive multilingual zero-shot TTS model supporting over 600 languages with SOTA performance, addressing the critical limitation of narrow language coverage in existing zero-shot TTS systems. The core of OmniVoice is a novel diffusion language model-style single-stage discrete-token NAR framework, which directly maps text to multi-codebook acoustic tokens to circumvent the inherent limitations of conventional two-stage models. To unlock the full potential of this streamlined architecture, we propose a full-codebook random masking strategy to enhance training efficiency and LLM weight initialization to resolve the intelligibility challenge. Trained on a 581k-hour multilingual dataset curated exclusively from open-source resources, OmniVoice achieves the broadest language coverage of zero-shot TTS models to date, delivers SOTA performance on Chinese, English and diverse multilingual benchmarks, exhibits exceptional generalization to low-resource languages, and is equipped with multi-dimensional controllability and high inference efficiency, marking a significant advance in expanding high-quality TTS capabilities to a broad spectrum of the world’s languages.

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A Complete List of Multilingual Training Data

We used the following datasets to train OmniVoice:

Emilia [46], Emilia-YODAS [46], LibriTTS [61], Common Voice [62], VoxBox [7], Meta Omnilingual ASR Corpus [57], FLEURS [52], GigaSpeech 2 [63], YODAS-Granary [64], EuroSpeech [65], IndicVoices-R [66], IndicTTS [67], Rasa [68], PhoAudiobook [69], viVoice², CML-TTS [70], Wenetspeech-yue [71], Wenetspeech-chuan [72], Kespeech [73], AESRC [74], Ksponspeech [75], Reazonspeech [76], AISHELL6-whisper [77], ParaSpeechCaps [78], Nvspeech [30], NonVerbalSpeech-38K [45], SeniorTalk [79], ChildMandarin [80], KSC2 [81], CIArTTS [82], ArVoice [83], KazakhTTS2 [84], RuLS³, IMUT-MC [85], HUI-Audio-Corpus-German⁴, MGB-2 [86], MGB-3 [87], MGB-5 [88], SADA [89], MASC [90], Mixat [91], Arabic datasets found from [37] (darija speech to text⁵, DarijaTTS-clean⁶, Jordan-Audio⁷, UAE 100K⁸), NICT-Tib1 [92], TIBMD@MUC [93], XBMU-AMDO3 [94], Tibetan-Greetings [95].

B Results of Different Inference Steps

In Table 9, we evaluate the multilingual version OmniVoice model with different inference steps on English and Chinese benchmarks.

Table 9: Objective evaluation results of OmniVoice with different inference steps.

Steps	LibriSpeech-PC test-clean			Seed-TTS test-en			Seed-TTS test-zh		
	SIM-o ↑	WER ↓	UTMOS ↑	SIM-o ↑	WER ↓	UTMOS ↑	SIM-o ↑	WER ↓	UTMOS ↑
64	0.729	1.28	4.30	0.742	1.60	3.92	0.777	0.81	3.13
32	0.729	1.30	4.28	0.741	1.53	3.91	0.777	0.84	3.11
16	0.728	1.50	4.23	0.735	1.72	3.92	0.773	0.99	3.00
8	0.713	2.02	4.02	0.716	1.94	3.59	0.756	1.58	2.72

²<https://huggingface.co/datasets/capleaf/viVoice>

³<https://www.openslr.org/96/>

⁴<https://huggingface.co/datasets/Paradoxia/opendata-iisys-hui>

⁵https://huggingface.co/datasets/adiren7/darija_speech_to_text

⁶<https://huggingface.co/datasets/KandirResearch/DarijaTTS-clean>

⁷<https://huggingface.co/datasets/nadsoft/Jordan-Audio>

⁸https://huggingface.co/datasets/AhmedBadawy11/UAE_100K

C Detailed Results on FLEURS-Multilingual-102

Per-language CERs of of ground-truth and OmniVoice on FLEURS-Multilingual-102 is shown in Fig. 4.

Table 10: Per-language CERs of ground-truth and OmniVoice on FLEURS-Multilingual-102

Language	Ground-truth	OmniVoice	Language	Ground-truth	OmniVoice
Afrikaans	3.54	3.08	Amharic	4.98	7.30
Armenian	1.38	1.45	Assamese	5.29	3.92
Asturian	3.79	2.07	Azerbaijani	2.60	3.20
Belarusian	1.80	1.00	Bengali	4.10	3.08
Bosnian	1.61	0.66	Bulgarian	1.66	0.87
Burmese	6.77	12.20	Cantonese	11.01	21.92
Catalan	1.17	0.86	Cebuano	1.74	1.81
Central Kurdish	3.92	2.97	Chichewa	4.63	3.32
Chinese	5.54	3.16	Croatian	4.64	0.93
Czech	2.19	1.21	Danish	2.86	1.79
Dutch	2.01	1.87	English	2.63	1.67
Estonian	0.97	1.03	Filipino	2.34	1.24
Finnish	1.60	1.46	French	2.13	1.95
Fulah	12.33	7.70	Galician	1.52	1.03
Ganda	7.87	6.60	Georgian	1.78	1.83
German	2.22	1.35	Greek	2.31	1.57
Gujarati	3.06	2.74	Hausa	5.14	3.03
Hebrew	4.91	9.75	Hindi	2.54	2.64
Hungarian	1.75	1.04	Icelandic	2.82	3.77
Igbo	7.75	7.40	Indonesian	1.87	2.95
Irish	17.56	8.52	Italian	1.27	1.56
Japanese	7.70	5.96	Javanese	3.17	2.80
Kabuverdianu	3.29	2.10	Kamba	8.81	10.42
Kannada	2.84	2.70	Kazakh	2.01	3.11
Khmer	6.20	11.48	Kirghiz	2.33	1.80
Korean	3.72	3.78	Lao	22.87	25.51
Latvian	1.87	1.55	Lingala	2.36	1.68
Lithuanian	2.98	2.05	Luo	3.58	3.37
Luxembourgish	5.48	4.52	Macedonian	1.05	0.89
Malay	1.59	1.19	Malayalam	2.99	3.40
Maltese	2.01	3.65	Maori	4.90	1.98
Marathi	5.02	2.76	Mongolian	4.78	4.26
Nepali	4.76	3.43	Norwegian Bokmål	1.50	1.14
Occitan	9.63	6.99	Odia	5.77	5.53
Oromo	12.50	4.97	Panjabi	5.76	2.95
Pedi	4.88	4.44	Persian	1.57	1.90
Polish	1.14	0.61	Portuguese	1.45	1.22
Pushto	11.42	5.53	Romanian	2.15	0.94
Russian	1.68	1.10	Serbian	1.36	0.82
Shona	4.07	1.80	Sindhi	5.51	3.26
Slovak	2.76	0.85	Slovenian	2.40	1.20
Somali	10.13	4.86	Spanish	1.08	0.77
Standard Arabic	2.58	1.92	Swahili	2.41	1.37
Swedish	2.14	1.43	Tajik	3.17	2.36
Tamil	4.00	3.77	Telugu	4.51	3.77
Thai	6.98	7.71	Turkish	1.94	2.71
Ukrainian	1.45	1.23	Umbundu	6.97	5.44
Urdu	80.27	28.73	Uzbek	3.31	2.23
Vietnamese	3.49	2.63	Welsh	3.91	3.37
Wolof	12.17	6.87	Xhosa	3.62	3.83
Yoruba	17.97	21.37	Zulu	3.33	2.03

D Supported Languages of OmniVoice

OmniVoice supports **646 languages** with a total of **581k hours** of training data. Detailed language, along with its OmniVoice language ID, ISO 639-3 code, and training data duration are shown in Table 11.

Table 11: Supported languages of OmniVoice along with its OmniVoice language ID, ISO 639-3 code, and training data duration (hours).

Language	ID	ISO	Hours	Language	ID	ISO	Hours	Language	ID	ISO	Hours
Abadi	kbt	kbt	9.73	Abkhazian	ab	abk	57.27	Abroon	abr	abr	9.22
Abua	abn	abn	10.27	Adamawa Fulfulde	fub	fub	13.12	Adyghe	ady	ady	32.6
Afada	aal	aal	10.19	Afrikaans	af	af	4.4	Agwagwune	yay	yay	8.26
Aja (Benin)	ajg	ajg	5.63	Akebu	keu	keu	9.1	Alago	ala	ala	11.04
Albanian	sq	sqi	8.59	Algerian Arabic	arq	arq	9.64	Algerian Saharan Arabic	aao	aao	2.02
Ambo-Pasco Quechua	qva	qva	9.59	Ambonese Malay	abs	abs	10.03	Amdo Tibetan	adx	adx	56.94
Amharic	am	amh	12.83	Anaang	anw	anw	9.65	Angika	anp	anp	10.65
Antankarana Malagasy	xmv	xmv	17.9	Aragonese	an	arg	16.4	Arbëreshë Albanian	aas	aas	6.11
Arequipa-La Unión Quechua	qxu	qxu	10.12	Armenian	hy	hye	42.15	Ashe	ahs	ahs	10.62
Ashéninka Perené	prq	prq	7.16	Askopan	eiv	eiv	10.44	Assamese	as	asm	270.85
Asturian	ast	ast	8.48	Atayal	tay	tay	7.02	Awak	awo	awo	10.22
Ayacucho Quechua	quy	quy	0.05	Azerbaijani	az	aze	9.84	Baatonum	bba	bba	10.53
Bacama	bcy	bcy	9.94	Bade	bde	bde	9.89	Bafia	bsf	bsf	16.43
Bafut	bfd	bfd	9.03	Bagirmi Fulfulde	fui	fui	15.04	Bago-Kusuntu	bkg	bkg	8.86
Baharna Arabic	abv	abv	10.41	Bakoko	bkh	bkh	6.0	Balanta-Ganja	bjt	bjt	9.41
Balti	bft	bft	16.28	Bamenyam	bce	bce	9.9	Bamun	bax	bax	10.24
Bangwinji	bsj	bsj	10.0	Banjar	bjn	bjn	11.68	Bankon	abb	abb	11.2
Baoulé	bci	bci	10.21	Bara Malagasy	bhr	bhr	12.14	Barok	bjk	bjk	10.16
Basa (Cameroon)	bas	bas	10.66	Basa (Nigeria)	bzw	bzw	10.27	Bashkir	ba	bak	249.1
Basque	eu	eus	479.86	Batak Mandailing	btm	btm	11.09	Batanga	bnm	bnm	15.01
Bateri	btv	btv	9.8	Bats	bbl	bbl	11.22	Bayot	bda	bda	9.47
Bebele	beb	beb	7.52	Belarusian	be	bel	1809.43	Bengali	bn	ben	271.76
Betawi	bew	bew	11.15	Bhili	bhb	bhb	9.98	Bhojpuri	bho	bho	10.05
Bilur	bxf	bxf	10.84	Bima	bhp	bhp	10.67	Bodo	brx	brx	231.57
Boghom	bux	bux	10.48	Bokyi	bky	bky	9.85	Bomu	bmq	bmq	10.68
Bondei	bou	bou	9.98	Borgu Fulfulde	fue	fue	20.1	Bosnian	bs	bos	690.73
Brahui	brh	brh	19.89	Braj	bra	bra	10.68	Breton	br	bre	25.48
Buduma	bdm	bdm	10.17	Buginese	bug	bug	11.09	Bukharic	bhh	bhh	11.38
Bulgarian	bg	bul	2190.76	Bulu (Cameroon)	bum	bum	9.06	Bundeli	bns	bns	10.88
Bunun	bnn	bnn	9.26	Bura-Pabir	bwr	bwr	10.4	Burak	bys	bys	9.92
Burmese	my	mya	12.14	Burushaski	bsk	bsk	9.14	Cacaloxtepec Mixtec	miu	miu	9.18
Cajatambo North Lima Quechua	qvl	qvl	9.95	Cakfem-Mushere	cky	cky	8.96	Cameroon Pidgin	wes	wes	10.06
Campidanese Sardinian	sro	sro	10.16	Cantonese	yue	yue	13302.38	Catalan	cat	cat	3358.6
Cebuano	ceb	ceb	12.17	Cen	cen	cen	9.85	Central Kurdish	ckb	ckb	137.52
Central Nahuatl	nhn	nhn	9.51	Central Pame	pbs	pbs	9.69	Central Pashto	pst	pst	11.4
Central Puebla Nahuatl	ncx	ncx	9.86	Central Tarahumara	tar	tar	9.73	Central Yupik	esu	esu	2.18
Central-Eastern Niger Fulfulde	fuq	fuq	9.28	Chadian Arabic	shu	shu	2.29	Chichewa	ny	nya	10.8
Chichicapan Zapotec	zpv	zpv	9.85	Chiga	egg	egg	10.84	Chimalapa Zoque	zoh	zoh	9.35
Chimborazo Highland Quichua	qug	qug	10.12	Chinese	zh	cmn	111343.3	Chiquián Ancash Quechua	qxa	qxa	9.99
Chitwania Tharu	the	the	10.06	Chokwe	cjk	cjk	11.01	Chuvash	chv	chv	23.96
Cibak	ckl	ckl	10.91	Coastal Konjo	kjc	kjc	10.18	Copainalá Zoque	zoc	zoc	10.07
Cornish	kw	cor	12.15	Corongo Ancash Quechua	qwa	qwa	9.72	Croatian	hr	hrv	2795.31
Cross River Mbembe	mfn	mfn	10.03	Cuyamecalco Mixtec	xtu	xtu	9.4	Czech	cs	ces	148.13
Dadiya	dbd	dbd	9.61	Dagbani	dag	dag	10.14	Dameli	dml	dml	9.18
Danish	da	dan	1665.98	Dargwa	dar	dar	1.22	Dazaga	dzg	dzg	9.96
Deccan	dcc	dcc	10.38	Degema	deg	deg	11.07	Dera (Nigeria)	kna	kna	11.91
Dghwede	dgh	dgh	9.95	Dhatki	mki	mki	8.83	Dhivehi	dv	div	38.61
Dhofari Arabic	adf	adf	0.31	Dijim-Bwilim	cfa	cfa	10.32	Dogri	dgo	dgo	117.04
Domaaki	dmk	dmk	6.38	Dotyali	dty	dty	10.85	Duala	dua	dua	12.14
Dutch	nl	nl	2264.13	Düya	ldb	ldb	11.31	Dyula	dyu	dyu	0.33
Eastern Balochi	bgp	bgp	10.98	Eastern Bolivian Guarani	gui	gui	22.72	Eastern Egyptian Bedawi Arabic	avl	avl	1.86
Eastern Krahn	kqo	kqo	9.28	Eastern Mari	mhr	mhr	272.31	Eastern Yiddish	ydd	ydd	18.43
Ebréi	ebr	ebr	1.5	Eggon	ego	ego	9.95	Egyptian Arabic	arz	arz	23.23
Ejagham	etu	etu	10.3	Elme	elm	elm	11.27	Eloyi	afo	afo	11.21
Embu	ebu	ebu	9.81	English	en	eng	206061.1	Erzya	myv	myv	3.1
Esan	ish	ish	10.05	Esperanto	eo	epo	1396.64	Estonian	et	est	960.37
Eton (Cameroon)	eto	eto	7.43	Ewondo	ewo	ewo	12.71	Extremaduran	ext	ext	13.59
Fang (Equatorial Guinea)	fan	fan	3.51	Fanti	fat	fat	11.38	Farefare	gur	gur	9.24
Fe'fe'	fmp	fmp	9.86	Filipino	fil	fil	7.71	Filomena Mata-Coahuatlán Totonac	tlp	tlp	11.35
Finnish	fi	fin	468.62	Fipa	fil	fil	10.55	French	fr	fra	23675.32
Fulah	ff	ful	13.84	Galician	gl	glg	208.81	Gambian Wolof	wof	wof	9.46
Ganda	lg	lug	447.82	Garhwali	gbm	gbm	19.14	Gawar-Bati	gwt	gwt	12.16
Gawri	gwc	gwc	10.83	Gbagyi	gbr	gbr	12.12	Gbari	gby	gby	12.59
Geji	gyz	gyz	10.49	Gen	gej	gej	5.39	Georgian	ka	kat	156.96
German	de	deu	21927.13	Geser-Gorom	ges	ges	10.08	Gheg Albanian	aln	aln	3.92
Ghomala'	bbj	bbj	7.32	Gidar	gid	gid	10.06	Glavda	glw	glw	10.51
Goan Konkani	gom	gom	9.82	Goaria	gig	gig	9.41	Goemai	ank	ank	10.0
Gola	gol	gol	9.26	Greek	el	ell	2412.54	Guarani	gn	grn	4.06
Guduf-Gava	gdf	gdf	12.21	Guerrero Amuzgo	amu	amu	10.1	Gujarati	gu	guj	91.18
Gujari	gju	gju	8.66	Gulf Arabic	afb	afb	98.55	Gurgula	ggg	ggg	7.12
Gusii	guz	guz	9.5	Gusilay	gsi	gsi	10.0	Gweno	gwe	gwe	8.87
Guilá Zapotec	ztu	ztu	9.17	Hadothi	hoj	hoj	10.08	Hahon	hah	hah	9.64
Haitian	ht	hat	0.04	Hakha Chin	cnh	cnh	2.24	Hakö	hao	hao	8.56
Halia	hla	hla	9.86	Hausa	ha	hau	17.75	Hawaiian	haw	haw	11.79
Hazaragi	haz	haz	9.69	Hebrew	he	heb	13.4	Hemba	hem	hem	9.53
Herero	hz	her	9.59	Highland Konjo	kjk	kjk	10.21	Hijazi Arabic	acw	acw	22.32
Hindi	hi	hin	117.17	Huarijio	var	var	9.28	Huautla Mazatec	mau	mau	6.39
Huasteca Nahuatl	nhq	nhq	5.07	Huba	hbb	hbb	10.7	Huitepec Mixtec	mxx	mxx	9.64
Hula	hul	hul	10.33	Hungarian	hu	hun	255.83	Hunjara-Kaina Ke	hkk	hkk	8.69
Hwana	hwo	hwo	11.23	Ibibio	ibb	ibb	7.38	Icelandic	is	isl	647.29
Idakho-Isukha-Tiriki	ida	ida	9.31	Idoma	idu	idu	11.16	Igbo	ibo	ibo	13.69
Igo	ahl	ahl	9.22	Ikposo	kpo	kpo	7.83	Ikweri	ikw	ikw	10.0
Imbabura Highland Quichua	qvi	qvi	11.0	Indonesian	id	ind	6327.87	Indus Kohistani	mvy	mvy	21.64
Interlingua	ia	ina	13.48	Inupiaq	ik	ikp	2.11	Irish	ga	gle	21.4
Iron Ossetic	os	oss	1.38	Isekiri	its	its	11.85	Isoko	iso	iso	10.33
Italian	it	ita	9402.46	Ito	itw	itw	9.19	Itzá	itz	itz	7.08
Ixtayutla Mixtec	vmj	vmj	10.17	Izon	ijc	ijc	9.95	Jambi Malay	jax	jax	10.29
Japanese	ja	jpn	36914.4	Jaquaru	jqr	jqr	9.32	Jauja Wanca Quechua	qxw	qxw	11.42
Jansari	jns	jns	11.25	Javanese	jv	jav	11.19	Jiba	juo	juo	10.43
Jju	kaj	kaj	10.16	Judeo-Moroccan Arabic	aju	aju	7.21	Juxtlahuaca Mixtec	vmc	vmc	9.43
Kabardian	kbd	kbd	108.35	Kabras	lkb	lkb	9.99	Kabuverdianu	kea	kea	10.51
Kabyle	kab	kab	529.52	Kachi Koli	gjk	gjk	20.83	Kairak	ckr	ckr	10.51

Language	ID	ISO	Hours	Language	ID	ISO	Hours	Language	ID	ISO	Hours
Kalabari	ijn	ijn	11.04	Kalasha	kls	kls	9.11	Kalenjin	klm	klm	40.42
Kalkoti	xka	xka	8.0	Kamba	kam	kam	14.72	Kamo	kcq	kcq	10.49
Kanauji	bij	bij	11.01	Kanembu	kbl	kbl	10.19	Kannada	kn	kan	128.06
Karekare	kai	kai	10.52	Kashmiri	ks	kaz	110.42	Kathoriya Tharu	ktk	ktk	10.64
Kati	bsh	bsh	8.77	Kazakh	ogo	ogo	1537.29	Keiyo	eyo	eyo	9.24
Khams Tibetan	khg	khg	6.38	Khana	khw	khw	15.55	Khetrani	xhe	xhe	9.4
Khmer	km	khm	7.1	Khowar	rw	kin	2021.66	Kinga	zga	zga	9.5
Kinnari	kfk	kfk	10.32	Kinyarwanda	thq	thq	10.28	Kirghiz	ky	kir	46.63
Kirya-Konzäl	fkx	fkx	9.98	Kochila Tharu	trp	trp	10.74	Kohistani Shina	plk	plk	12.75
Kohumono	bes	bes	10.45	Kok Borok	kmy	kmy	10.28	Kol (Papua New Guinea)	kol	kol	9.95
Kom (Cameroon)	koo	koo	13.23	Koma	ko	kor	8609.28	Konkani	knn	knn	112.83
Konzo	kfe	kfe	10.25	Koti	eko	eko	8.15	Korwa	kfp	kfp	11.87
Kuanama	kj	kua	9.88	Kui (India)	uki	uki	10.77	Kuanua	ksd	ksd	9.91
Kuot	kto	kto	9.77	Kushi	kuh	kuh	10.35	Kulung (Nigeria)	bbu	bbu	10.39
Kwasio	nmg	nmg	10.39	Lala-Roba	lla	lla	10.38	Kwambi	kwm	kwm	9.9
Lao	lo	lo	7.63	Larike-Wakasihu	alo	alo	9.97	Lamang	hia	hia	11.07
Latgalian	ltg	ltg	27.23	Latvian	lv	lav	1441.58	Lasi	lss	lss	6.53
Liana-Seti	ste	ste	10.43	Liberia Kpelle	xpe	xpe	9.5	Levantine Arabic	apc	apc	15.65
Libyan Arabic	ayl	ayl	20.13	Ligurian	lij	lij	15.97	Liberian English	lir	lir	10.26
Lingala	ln	lin	17.99	Lithuanian	lit	lit	2629.45	Lijili	lgi	lgi	10.89
Logooli	rag	rag	9.39	Logudorese Sardinian	src	src	10.67	Lokari	lrk	lrk	10.5
Loloda	loa	loa	9.31	Longuda	lnu	lnu	10.46	Loja Highland Quichua	qvj	qvj	10.59
Luba-Lulua	lua	lua	8.47	Luo	luo	luo	36.17	Loxicha Zapotec	ztp	ztp	9.62
Luxembourgish	lb	ltz	8.46	Maasina Fulfulde	ffm	ffm	10.46	Lushai	lus	lus	20.24
Macedo-Romanian	rup	rup	0.02	Macedonian	mki	mki	27.21	Maba (Chad)	mde	mde	9.5
Mafa	maf	maf	9.97	Maithili	mai	mai	131.37	Mada (Cameroon)	mxu	mxu	12.0
Malayalam	ml	mal	166.57	Mali	gcc	gcc	9.87	Malay	ms	msa	9.57
Maltese	mt	mlt	630.29	Mandara	tbf	tbf	10.01	Malinaltepec Me'phaa	tcf	tcf	9.04
Mangarai	mgy	mgy	10.5	Manipuri	mni	mni	44.46	Mandjak	mfv	mfv	9.55
Manx	gv	glv	10.07	Maori	mi	mri	18.02	Mansoanka	msw	msw	9.32
Marghi Central	mrt	mrt	10.36	Marghi South	mfm	mfm	10.05	Marathi	mr	mar	156.71
Marwari (Pakistan)	mve	mve	9.96	Masana	mcn	mcn	10.09	Maria (India)	mrr	mrr	11.0
Matsés	mcf	mcf	9.61	Mazaltepec Zapotec	mzo	mzo	10.09	Masikoro Malagasy	msh	msh	14.16
Mazatlán Mixe	mzl	mzl	10.05	Mbe	mfo	mfo	10.24	Mazatlán Mazatec	vmz	vmz	9.82
Mbum	mdl	mdl	9.82	Medumba	byv	byv	10.95	Mbo (Cameroon)	mbo	mbo	9.51
Meru	mer	mer	9.89	Mesopotamian Arabic	acm	acm	3.78	Mekeo	mek	mek	9.18
Min Nan Chinese	nan	nan	17.55	Mingrelian	xmf	xmf	11.47	Mewari	mtr	mtr	10.58
Miya	mkf	mkf	10.16	Mokpwe	bri	bri	7.53	Mitlatongo Mixtec	vmm	vmm	9.95
Mom Jango	ver	ver	10.93	Mongolian	mn	mon	269.08	Moksha	mdf	mdf	0.47
Motu	meu	meu	9.88	Mungaka	mxh	mxh	7.53	Moroccan Arabic	ary	ary	104.67
Mundang	mua	mua	9.2	Musi	mui	mui	10.52	Mpumpong	mng	mng	4.94
Najdi Arabic	ars	ars	203.54	Nalik	nal	nal	10.33	Musey	mse	mse	7.21
Ndonga	ng	ndo	9.08	Neapolitan	nap	nap	9.97	Naba	mne	mne	10.37
Ngamo	nbh	nbh	10.04	Ngas	anc	anc	10.14	Nawdm	nmz	nmz	6.3
Ngizim	ngi	ngi	10.06	Ngomba	jgo	jgo	10.15	Nepali	npi	npi	171.5
Nigerian Fulfulde	fuv	fuv	9.97	Nigerian Pidgin	pcm	pcm	11.04	Ngiemboon	nnh	nnh	16.15
Nobiin	fia	fia	9.96	North Mesopotamian Arabic	ayp	ayp	10.92	Ngombale	nla	nla	8.79
Northern Betsimisaraka Malagasy	bmm	bmm	19.12	Northern Pashto	hno	hno	20.04	Nimadi	noe	noe	11.12
Northern Pame	pmq	pmq	10.24	Norwegian	nbu	nbu	11.03	North Moluccan Malay	max	max	9.43
Northwest Gbaya	gya	gya	8.45	Noti	ncf	ncf	9.84	Northern Kurdish	kmr	kmr	69.59
Norwegian Nynorsk	nn	nno	1.54	Nzanyi	nja	nja	10.02	Northern Uzbek	uzn	uzn	15.23
Nyungwe	nyu	nyu	8.98	Od	odk	odk	20.26	Norwegian Bokmål	nb	nob	12.7
Occitan	oci	oci	16.8	Omani Arabic	acx	acx	22.03	Nyankpa	yes	yes	10.26
Oduai	odu	odu	10.57	Ormuri	oru	oru	16.74	Nüpode Huitoto	hux	hux	9.04
Orma	orc	orc	22.01	Paiwan	pwn	pwn	13.76	Odia	ory	ory	144.81
Pahari-Potwari	phr	phr	24.03	Parkari Koli	kvx	kvx	11.04	Orizaba Nahuatl	nlv	nlv	11.42
Papuan Malay	pmy	pmy	10.17	Persian	fa	fars	366.07	Oromo	om	orm	6.6
Pero	phl	phl	20.69	Piemontese	pms	pms	16.01	Panjabi	pa	pan	147.37
Phalura	plt	plt	19.39	Polish	pl	pol	911.68	Pedi	nso	nso	12.64
Plateau Malagasy	pt	por	16855.05	Pulaar	fuc	fuc	14.77	Petats	pex	pex	10.2
Portuguese	qxp	qxp	9.81	Pushto	ps	pus	88.62	Piya-Kwonci	piy	piy	10.38
Puno Quechua	byx	byx	9.79	Quiotepec Chinantec	chy	chy	9.76	Pogomam	poc	poc	9.63
Qaqet	lag	lag	9.47	Rapoisi	kyx	kyx	9.17	Pular	fuf	fuf	13.77
Rangi	zor	zor	9.04	Romanian	ro	ron	70.23	Pükoot	pko	pko	10.4
Rayón Zoque	rof	rof	18.9	Rotokas	roo	roo	9.07	Rana Tharu	rth	rth	9.99
Rombo	ru	rus	20338.5	Sacapulteco	quv	quv	8.9	Ratahan	rhm	rhm	9.34
Sakalava Malagasy	skg	skg	9.02	Sakizaya	szs	szs	11.47	Romansh	roh	roh	9.21
Samba Daka	ccg	ccg	10.11	Samba Leko	ndi	ndi	11.27	Rukai	dru	dru	9.26
San Francisco Del Mar Huave	hue	hue	9.45	San Juan Atzingo Popoloca	poe	poe	10.01	Saïdi Arabic	aec	aec	9.28
San Miguel El Grande Mixtec	mig	mig	9.66	Sansi	ssi	ssi	10.47	Saleman	sau	sau	10.53
Santa Ana de Tusi Pasco Quechua	qxt	qxt	10.05	Santa Catarina Albarradas Zapotec	ztn	ztn	10.02	San Felipe Otaltepec Popoloca	pow	pow	8.84
Santiago del Estero Quichua	qus	qus	9.55	Saposa	spz	spz	9.81	San Martín Itunyoso Triqui	trq	trq	8.24
Sardinian	sc	srđ	2.77	Saya	say	say	10.02	Sanskrit	sa	san	84.49
Serbian	sr	srp	1855.33	Seri	sei	sei	9.81	Santali	sat	sat	98.37
Shona	sn	sna	9.96	Siar-Lak	sir	sir	9.87	Saraiki	skr	skr	4.13
Sicilian	scn	scn	13.35	Sihuas Ancash Quechua	qws	qws	10.18	Sediq	trv	trv	7.77
Sinaugoro	snc	snc	10.38	Sindhi	sd	snd	46.27	Shina	scl	scl	9.84
Sinhala	si	sin	11.98	Sincahua Mixtec	xti	xti	9.5	Sibe	nco	nco	9.96
Siwai	siw	siw	10.47	Slovak	sk	slk	2478.46	Sikkimese	sip	sip	10.07
Solos	sol	sol	9.95	Somali	so	som	13.22	Sindhi Bhil	sbm	sbm	10.53
South Giziga	giz	giz	10.03	South Ucyali Ashéninka	cpy	cpy	9.15	Sipacapense	qum	qum	9.37
Southern Betsimisaraka Malagasy	bzc	bzc	17.45	Spanish	es	spa	2759.74	Slovenian	sl	slv	1172.61
Soyaltepec Mazatec	vmp	vmp	10.17	Sudanese Arabic	apd	apd	9.93	Soninke	snk	snk	10.04
Standard Moroccan Tamazight	zgh	zgh	1.19	Swahili	sw	swa	418.41	Southeastern Nochixtlán Mixtec	mxy	mxy	9.48
Svan	sva	sva	15.11	Tahaggart Tamahaq	thv	thv	4.25	Southern Pastaza Quechua	qup	qup	11.13
Tae'	rob	rob	9.02	Tamil	ta	tam	423.09	Standard Arabic	arb	arb	1483.53
Tajik	tg	tgk	9.23	Tanosy Malagasy	txy	txy	12.07	Sulka	sua	sua	10.12
Tangale	tan	tan	10.14	Tedaga	tuq	tuq	10.0	Swedish	sv	swe	2453.14
Tatar	tt	tat	30.03	Tecop	tio	tio	9.85	Taita	dav	dav	9.12
Tem	kdh	kdh	4.07	Tera	trr	trr	9.89	Tandroy-Mahafaly Malagasy	tdx	tdx	3.81
Tepinapa Chinantec	cte	cte	9.54	Tesaka Malagasy	tkg	tkg	17.86	Tarok	yer	yer	10.08
Teranu	twu	twu	11.45	Thai	th	tha	10499.77	Telugu	tel	tel	230.21
Teutla Cuicatec	cut	cut	8.04	Tidore	tvo	tvo	10.31	Tepeuxila Cuicatec	cux	cux	7.83
Tidaá Mixtec	mtx	mtx	9.09	Tigrinya	tir	tir	0.08	Terei	buo	buo	9.48
Tigre	tig	tig	7.49	Tlacoapa Me'phaa	tpl	tpl	9.28	Tetelcingo Nahuatl	nhg	nhg	8.92
Tinuput	tpz	tpz	9.33	Toki Pona	tok	tok	13.51	Tibetan	bo	bod	82.27
Tlingit	tli	tli	0.41	Tonsea	txs	txs	9.32	Tigak	tgk	tgk	9.71
Tondano	tdn	tdn	9.14	Torwali	trw	trw	14.98	Tilquapan Zapotec	zts	zts	9.33
Torau	ttu	ttu	9.87	Tswana	tn	tsn	4.24	Tlacoatzintepec Chinantec	ctl	ctl	10.04
Tsotso	lto	lto	9.77	Tula	tul	tul	9.79	Tomoip	tqp	tqp	10.1
Tuki	bag	bag	10.97	Tungva	lcm	lcm	9.77	Tooro	tj	tj	10.31
Tunen	tvu	tvu	9.85	Turkana	tuv	tuv	10.17	Tsimihety Malagasy	xmw	xmw	11.53
Tupuri	tui	tui	9.26	Tututepec Mixtec	mtu	mtu	10.13	Tugen	tuy	tuy	8.79
Turkmen	tk	tuk	2.86					Tulu	tcy	tcy	11.72
								Tunisian Arabic	aeb	aeb	21.63
								Turkish	tr	tur	125.36
								Twí	tw	twi	0.25

Language	ID	ISO	Hours	Language	ID	ISO	Hours	Language	ID	ISO	Hours
Ubaghara	byc	byc	11.11	Uighur	ug	uig	428.77	Ukrainian	uk	ukr	1851.97
Umbundu	umb	umb	10.59	Upper Sorbian	hsb	hsb	2.71	Urdu	ur	urd	211.27
Ushojo	ush	ush	6.36	Uzbek	uz	uzb	115.28	Vai	vai	vai	8.76
Vietnamese	vi	vie	8481.98	Votic	vot	vot	0.1	Vairo	vro	vro	15.66
Waci Gbe	wci	wci	8.02	Wadiyara Koli	kxp	kxp	20.0	Waja	wja	wja	10.22
Wakhi	wbl	wbl	11.67	Wanga	lwg	lwg	9.36	Wapan	juk	juk	10.22
Warji	wji	wji	11.39	Welsh	cy	cym	131.21	Wemale	weo	weo	9.09
Western Frisian	fy	fry	70.41	Western Highland Purepecha	pua	pua	10.17	Western Juxtlahuaca Mixtec	jmx	jmx	10.01
Western Maninkakan	mlq	mlq	9.83	Western Mari	mrj	mrj	32.26	Western Niger Fulfulde	fuh	fuh	9.69
Western Panjabi	pnb	pnb	10.0	Wolof	wo	wol	8.71	Wuzlam	udl	udl	9.23
Xanagüfa Zapotec	ztg	ztg	9.86	Xhosa	xh	xho	13.35	Yace	ekr	ekr	10.76
Yakut	sah	sah	16.08	Yalahatan	jal	jal	11.18	Yanahuanca Pasco Quechua	qur	qur	9.95
Yangben	yav	yav	8.7	Yaqui	yaq	yaq	9.93	Yauyos Quechua	qux	qux	9.35
Yekhee	ets	ets	10.11	Yiddish	yi	yid	1.81	Yidgha	ydq	ydq	9.89
Yoruba	yo	yor	15.66	Yutanduchi Mixtec	mab	mab	9.26	Zacatlán-Ahuacatlán-Tepetzintla Nahuatl	nhi	nhi	0.05
Zarma	dje	dje	10.72	Zaza	zza	zza	1.52	Zulu	zu	zul	14.83
Ömie	aom	aom	8.19								

E Limitations and Future Works

OmniVoice is currently trained solely on publicly available open-source datasets. Due to inconsistent annotation quality and uneven acoustic quality across these sources, substantial room remains for performance improvement when trained on more curated, high-quality data. Likewise, the scope and flexibility of the model’s instruction-following capacity are constrained by the diversity and quality of existing instruction-tuning data. Constructing high-quality instruction datasets would therefore notably enhance the model’s voice design and customization capabilities. Furthermore, OmniVoice has not yet been optimized for processing complex numeric sequences or mathematical patterns. Integrating an external text-normalization front-end would effectively strengthen the model’s performance in such scenarios. Additionally, unlike continuous-space NAR TTS models, where inference steps can be drastically reduced via techniques such as flow distillation [16], no existing approach enables comparable inference acceleration for discrete-space NAR TTS models. Exploring optimization strategies to decrease inference steps for this branch of models thus constitutes a promising direction for future work.